DAT.co mNAV Monitor

Selected Indicator: Modified Net Asset Value (mNAV)

Focus Company: MicroStrategy, Inc. (MSTR)

Deployed URL:
https://bitcoin-hw2-ibtuj51v9-shutton921205-8714s-projects.vercel.app/

GitHub Repository:
https://github.com/ShuttonChen/datco-mnav-dashboard

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Platform Vercel

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1. Selected Indicator

1.1. What Indicator Was Chosen?

The selected indicator is the **Modified Net Asset Value (mNAV)** multiple, also referred to as the **Premium to NAV** ratio. It quantifies how much the public equity market values a Digital Asset Treasury company (DAT.co) relative to the fair market value of its Bitcoin holdings.

Formula

$$\text{mNAV} = \frac{\text{Company Market Capitalization}}{\text{BTC Holdings} \times \text{Current BTC Price}}$$

where Market Cap = Share Price × Shares Outstanding.

The interpretation of mNAV is straightforward:

Table 1: mNAV Interpretation Guide

mNAV Value	Status	Meaning
> 1.0	Premium	Market values the company above its BTC NAV — investors pay extra for Bitcoin exposure through this vehicle
$= 1.0$	Parity	Stock trades at exact Bitcoin NAV — no premium or discount
< 1.0	Discount	Market values the company below BTC NAV — rare, indicates fear of insolvency or forced liquidation risk

1.2. Why mNAV Was Chosen

mNAV was selected for the following reasons:

1. **Direct Bitcoin linkage.** Unlike most equity metrics, mNAV is mechanically tied to Bitcoin price, creating a live bridge between crypto and traditional stock markets.

2. **Institutional sentiment proxy.** A high mNAV (e.g., 3×) reveals that investors are *willing to pay a premium* to hold BTC through a corporate structure — a powerful signal of institutional bullishness.

3. **Reflexivity dynamics.** When mNAV is elevated, DAT.co companies can issue equity at premium to purchase additional Bitcoin, which supports BTC price — forming a feedback loop that is unique to this asset class.

4. **Data availability.** MicroStrategy (MSTR) discloses its BTC holdings quarterly, and stock/BTC prices are freely available via Yahoo Finance and CoinGecko APIs, making this indicator fully reproducible.

5. **Historical richness.** MSTR's mNAV has ranged from $\approx 1.0\times$ (bear market floor) to $\approx 3.5\times$ (bull market peak), providing a meaningful time-series to analyze.

2. Relationship with Bitcoin (BTC)

2.1. Mechanical Relationship

The mNAV multiple is *directly computed* from the Bitcoin price. Holding all else constant:

$$\Delta\,\text{mNAV} \approx -\frac{\text{Market Cap}}{(\text{BTC} \times P_{\text{BTC}})^2} \cdot \text{BTC} \cdot \Delta P_{\text{BTC}}$$

That is, when BTC price rises and the stock price does *not* keep pace, mNAV compresses. Conversely, when MSTR stock rallies faster than BTC (as often happens in bull markets due to leverage and sentiment), mNAV expands.

2.2. Empirical Observations

Historical data on MicroStrategy reveals several important behavioral patterns:

Table 2: MSTR mNAV Behavior Across Market Phases

Market Phase	**BTC Trend**	**mNAV Range**	**Explanation**
Bull Market (2020–21)	Strong uptrend	2.5×–3.5×	Equity premium expands with risk appetite
Bear Market (2022)	Sharp decline	1.0×–1.5×	Premium collapses; liquidation fears emerge
Recovery (2023–24)	Gradual rise	1.5×–2.5×	Sentiment rebuilds; premium re-expands
Institutional Wave (2024–25)	New ATH	2.0×–3.0×	ETF approval drives institutional demand

2.3. Hypotheses on BTC Price Behavior

Based on mNAV dynamics, the following hypotheses can be formulated:

Hypothesis 1 — mNAV as a Sentiment Amplifier

DAT.co stocks such as MSTR exhibit **amplified beta** to Bitcoin. During bull markets, mNAV expands because equity investors pay a premium for leveraged Bitcoin exposure. During bear markets, mNAV collapses faster than BTC itself, reflecting deleveraging risk.

$\Rightarrow$ A rapidly expanding mNAV ($> 3\times$) may indicate **overheated sentiment** and could serve as a contrarian signal for near-term BTC price tops.

Hypothesis 2 — mNAV Compression as a BTC Buy Signal

When mNAV approaches 1.0×, the implicit message is that investors value MSTR's stock at *no premium* over its raw Bitcoin holdings. Historically, these episodes have coincided with BTC price troughs and have preceded strong recoveries.

⇒ mNAV near parity (1.0×–1.3×) has historically represented **attractive entry points** for both MSTR stock and BTC.

Hypothesis 3 — The Reflexivity Loop

When mNAV is high, MSTR can issue new equity at a premium to NAV, use the proceeds to purchase more Bitcoin, and thereby increase its Bitcoin-per-share metric. This capital structure arbitrage:

1. Supports BTC demand directly (MSTR is a marginal buyer at scale)
2. Signals confidence to other institutions
3. Further expands mNAV through sentiment feedback

⇒ High mNAV periods can become **self-reinforcing** in bull markets, creating structural upward pressure on BTC price.

3. Data Collection

3.1. Data Sources

Table 3: Data Sources Used in the Platform

Data	Source	Method	Update Freq.
BTC Spot Price	CoinGecko Public API	REST API (free)	Real-time (60s)
BTC 24h Change	CoinGecko	Same endpoint	Real-time
BTC Historical	CoinGecko `/market_chart`	REST API	Daily candles
MSTR Stock Price	Yahoo Finance v8	Via CORS proxy	Real-time (60s)
BTC Holdings	MicroStrategy IR filings	Static (manually updated)	Quarterly
Other DAT.co	Public company filings	Static snapshot	Quarterly

3.2. Key Parameters

As of the time of deployment, the following parameters were used for MicroStrategy:

- **BTC Holdings:** 528,185 BTC (per official Q1 2025 disclosure)

- **Shares Outstanding:** $\approx$ 240,000,000 shares
- **mNAV computed:** dynamically in the browser using live API data

4. Data Visualization & Website Features

4.1. Deployed Website

The platform is deployed at:

https://bitcoin-hw2-ibtuj51v9-shutton921205-8714s-projects.vercel.app/

Source code is available at:

https://github.com/ShuttonChen/datco-mnav-dashboard

4.2. User Interface Features

The dashboard provides the following visualizations and tools:

1. **Live KPI Cards.** Four real-time metric cards display:
   - Current MSTR mNAV multiple (with threshold indicator)
   - Bitcoin spot price and 24-hour change
   - MSTR stock price and 24-hour change
   - Total BTC holdings market value in USD

   Cards auto-refresh every 60 seconds.

2. **Historical mNAV Chart.** An interactive time-series chart built with Chart.js displays:
   - mNAV multiple over time (gold line)
   - Bitcoin price normalized overlay (blue dashed line)
   - Reference lines at 1.0× (parity) and 2.0× (historical average)
   - Period selector: 30D, 90D, 6M, 1Y

3. **DAT.co Company Comparison Table.** A snapshot table comparing mNAV across multiple Digital Asset Treasury companies including MSTR, MARA, RIOT, COIN, GLXY, and HUT.

4. **Educational Formula Panel.** Explains the mNAV formula, interpretation guide, and the relationship with Bitcoin sentiment.

5. **AI-Powered Analysis (Bonus).** Integration with the Claude API (`claude-sonnet-4-20250514`) to generate on-demand market analysis including trend interpretation, historical context, and BTC price implications.

4.3. Technical Architecture

Listing 1: Core mNAV Computation

```
// Core mNAV formula
function computeMNAV(mstrPrice, sharesOutstanding, btcHeld,
   btcPrice) {
    const marketCap = mstrPrice * sharesOutstanding;
    const btcNAV    = btcHeld * btcPrice;
    return marketCap / btcNAV;   // mNAV multiple
}

// Example: MSTR with 528,185 BTC
// mstrPrice = $385, shares = 240M, btcPrice = $97,500
// marketCap = $385    240,000,000 = $92.4B
// btcNAV    = 528,185    $97,500  = $51.5B
// mNAV      = $92.4B / $51.5B       1.79
```

Listing 2: Deployment Commands

```
# Install Vercel CLI
npm install -g vercel

# Deploy to production
cd datco-mnav-dashboard
vercel --prod

# Push to GitHub
git add .
git commit -m "Initial commit: DAT.co mNAV dashboard"
git push -u origin main
```

5. AI-Generated Summary (Bonus)

The platform integrates the **Anthropic Claude API** (`claude-sonnet-4-20250514`) to provide on-demand market analysis. When the user clicks "Generate Analysis," the following prompt is constructed dynamically from live market data and sent to the API:

> *"You are a financial analyst specializing in Digital Asset Treasury companies. Given current BTC price, MSTR stock price, and mNAV of [X]x, provide a 3-paragraph analysis covering: (1) mNAV interpretation and historical context, (2) relationship with Bitcoin sentiment, (3) key risk factors for investors to monitor."*

The response is streamed character-by-character to the UI to create a live-typing effect, making the analysis feel real-time and engaging. The feature uses the Anthropic `/v1/messages` endpoint directly from the browser.

6. Deployment

6.1. Platform: Vercel

The website is deployed as a **static single-file application** on Vercel's free tier. The entire application is contained in a single `index.html` file with no build step required.

Table 4: Deployment Configuration

Property	Value
Platform	Vercel (Free Tier)
Build Type	Static (no framework)
Output Directory	`./` (root)
Auto-deploy	GitHub integration
SSL	Automatic (HTTPS)
CDN	Vercel Edge Network (global)

6.2. Live URLs

- **Deployed Website:**
  https://bitcoin-hw2-ibtuj51v9-shutton921205-8714s-projects.vercel.app/
- **GitHub Source Code:**
  https://github.com/ShuttonChen/datco-mnav-dashboard

7. Conclusion

This project successfully implements a real-time web dashboard for monitoring the mNAV (Modified Net Asset Value) multiple of Digital Asset Treasury companies, with a primary focus on MicroStrategy (MSTR).

The mNAV indicator was demonstrated to be a powerful lens for understanding institutional Bitcoin sentiment. Key findings include:

- mNAV functions as a **sentiment amplifier** — it expands faster than BTC in bull markets and compresses faster in bear markets due to leverage and institutional reflexivity.
- Historical mNAV patterns suggest that values near 1.0× have coincided with BTC price troughs, while values above 3.0× have often preceded corrections.
- The reflexivity loop — where high mNAV enables equity issuance to buy more BTC — makes DAT.co companies **structural bulls** on Bitcoin price, creating a feedback dynamic that distinguishes this indicator from traditional equity multiples.

The platform achieves all required objectives: data collection via public APIs, daily-frequency time-series visualization, multi-company comparison, and AI-assisted interpretation via Claude API integration.

Disclaimer: This project is for educational purposes only. All data is approximated and should be verified with official sources. Nothing in this report or the deployed website constitutes financial advice.

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